School of Future Tech

Case Study Report

On

<u>Point of Sale (POS) Billing & GST System</u>

(PS-89)

by

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1. Introduction

The Point of Sale (POS) Billing & GST System is a software solution developed to simplify the billing process in retail stores. In many small shops, billing and tax calculations are still done manually, which can lead to errors and inefficiencies.

This project aims to automate the billing process by allowing users to select products, calculate GST automatically, and generate a bill for the customer. The system also keeps track of total sales and GST collected, helping store owners manage their daily transactions more effectively.

The project is implemented using **C++ programming language** and runs in a console-based environment.

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2. Problem Statement

Retail billing requires accurate pricing, tax calculation, and invoice generation. Manual billing systems are time-consuming and prone to human errors, especially when calculating taxes such as GST.

The goal of this project is to create a **Point of Sale (POS) Billing System** that automates product billing, performs GST calculations automatically, and generates a clear invoice for customers.

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3. Objectives

The main objectives of this project are:

- To automate the retail billing process
- To calculate GST accurately for different products
- To generate an invoice for the customer
- To maintain a daily sales summary
- To reduce human errors in manual billing

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4. System Features

The POS Billing & GST System provides the following features:

Product Selection

Users can select products using a **Product ID**, which simulates product scanning in real POS machines.

GST Calculation

The system automatically calculates the GST amount based on the product's GST percentage.

Bill Generation

After selecting products and quantities, the system generates a final invoice that displays:

- Subtotal
- GST amount
- Total bill amount

Daily Sales Summary

The system also displays the total sales and total GST collected for the day.

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5. Working

The working of the POS Billing System follows these steps:

1. The system stores product details such as **Product ID, Name, Price, and GST rate**.
2. The user enters the **Product ID** to select a product.
3. The system asks for the **quantity** of the product.
4. It calculates:
    - Product subtotal
    - GST amount
    - Total amount
5. The user can continue adding products.
6. When finished, the system generates the **final invoice**.
7. The system also displays the **daily sales summary**.

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6. Technologies Used

The following technologies and programming concepts were used:

- **Programming Language:** C++
- **Concepts Used:**
    - Structures (struct)
    - Arrays
    - Loops
    - Conditional statements
    - Input/Output formatting (iomanip)

These concepts help organize product data and perform calculations efficiently.

7.Output

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===== POS BILLING SYSTEM =====

Available Products:
ID   Name           Price     GST%
1    Milk           50        5
2    Bread          40        5
3    Chocolate      100       12
4    Shampoo        200       18
5    Soap           60        12

Enter Product ID (0 to finish): 2
Enter Quantity: 3
Added to cart: Bread
Price: 120
GST: 6

Available Products:
ID   Name           Price     GST%
1    Milk           50        5
2    Bread          40        5
3    Chocolate      100       12
4    Shampoo        200       18
5    Soap           60        12

Enter Product ID (0 to finish): 1
Enter Quantity: 4
Added to cart: Milk
Price: 200
GST: 10

Available Products:
ID   Name           Price     GST%
1    Milk           50        5
2    Bread          40        5
3    Chocolate      100       12
4    Shampoo        200       18
5    Soap           60        12

Enter Product ID (0 to finish): 0


========= INVOICE =========
Subtotal: Rs 320.00
Total GST: Rs 16.00
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TOTAL AMOUNT: Rs 336.00

Thank You! Visit Again!

====== DAILY SALES SUMMARY ======
Total Sales: Rs 336.00
GST Collected: Rs 16.00
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8. Advantages

- Reduces manual calculation errors
- Faster billing process
- Accurate GST calculation
- Easy to use

- Improves efficiency in retail billing

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9. Limitations

- Console-based interface
- Limited number of products
- Does not store data permanently
- No barcode scanning feature

These limitations can be improved in future versions.

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10. Future Enhancements

The system can be improved by adding the following features:

- Barcode scanning for real product scanning
- Database storage for products
- GUI-based interface
- Customer receipt printing
- Inventory management system

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11. Conclusion

The POS Billing & GST System successfully demonstrates how programming can automate retail billing operations. The system allows users to select products, calculate GST automatically, and generate a bill for the customer.

This project highlights the practical application of C++ programming concepts such as structures, loops, and conditional statements in solving real-world problems like retail billing.